

Strategic A.I. Implementation at Thomas Cook

December 2nd, 2024





CLIENT

Thomas Cook aims to enter the U.S. and establish itself into the travel & tourism Industry



OBJECTIVE

To propose strategies to integrate Artificial Intelligence (AI) at Thomas Cook to support its entry in the U.S. market.



DELIVERABLES

Delivering tailored, practical, and strategic solutions through competitor and user analysis, and technology assessment to meet client needs and drive company's growth and expansion

The Evolution of Thomas Cook



1841

Founded

Thomas Cook revolutionized modern tourism and travel. The company grew into a global leader offering diverse holiday packages

2019

Challenges in 2019

Thomas Cook faced significant operational and financial difficulties, leading to its collapse in September 2019.

2019

Revival and Innovation

Revived two months later by former employees, blending its rich legacy with modern technology to create a new venture.

2024

Problem Statement

After establishing a strong presence in Europe, Thomas Cook is now targeting the U.S., one of the most competitive travel markets. The company aims to integrate AI into its expansion, focusing on strategic recommendations, market analysis, and growth opportunities.





S.I.D.E Strategy

Proposed AI Technologies



S



AI-Driven Sentiment Analysis

Real-time customer feedback analysis for proactive service improvements.

I



Personalized Itinerary Planner

AI-powered assistant for bookings, FAQs, and tailored travel plans.

D



Predictive Demand and Dynamic Pricing

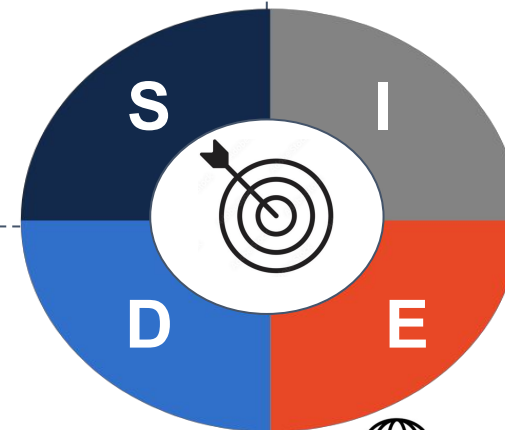
AI analysis of data to optimize pricing and maximize revenue.

E



Predictive Emergency and Health Alerts

AI-powered system for traveler safety and health monitoring.



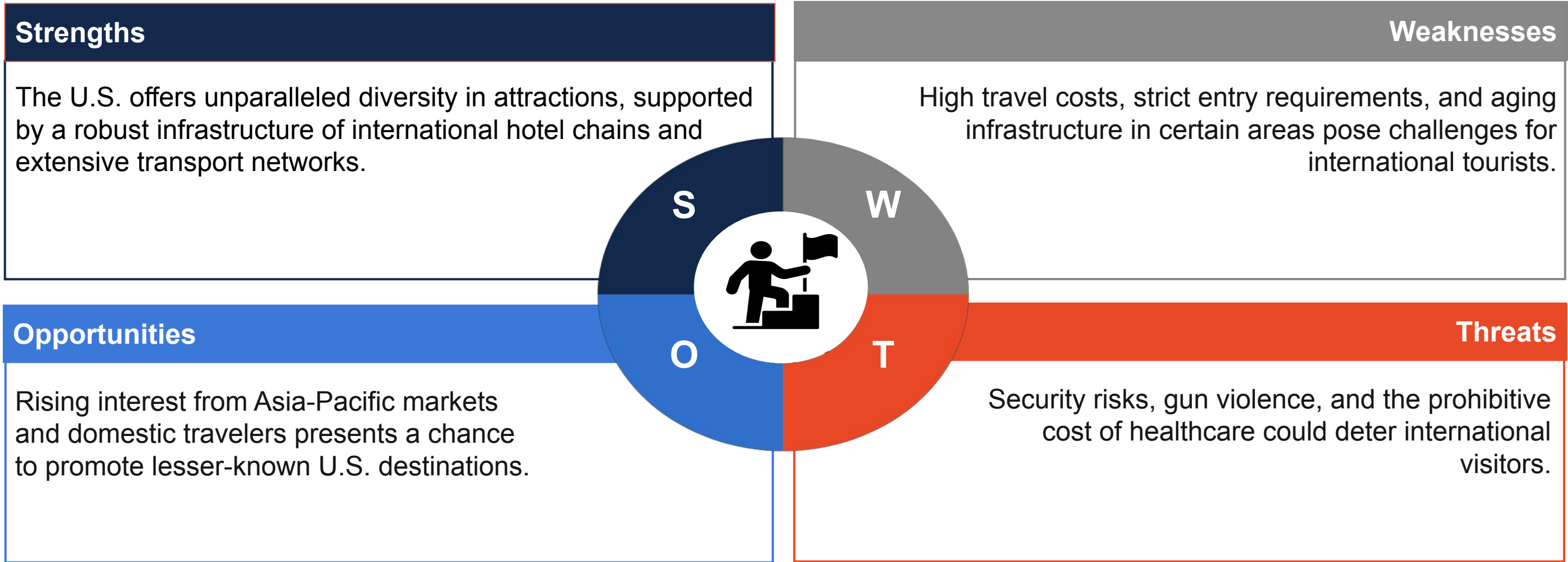


Research & Analysis

SWOT Analysis for U.S. Market



SWOT Analysis



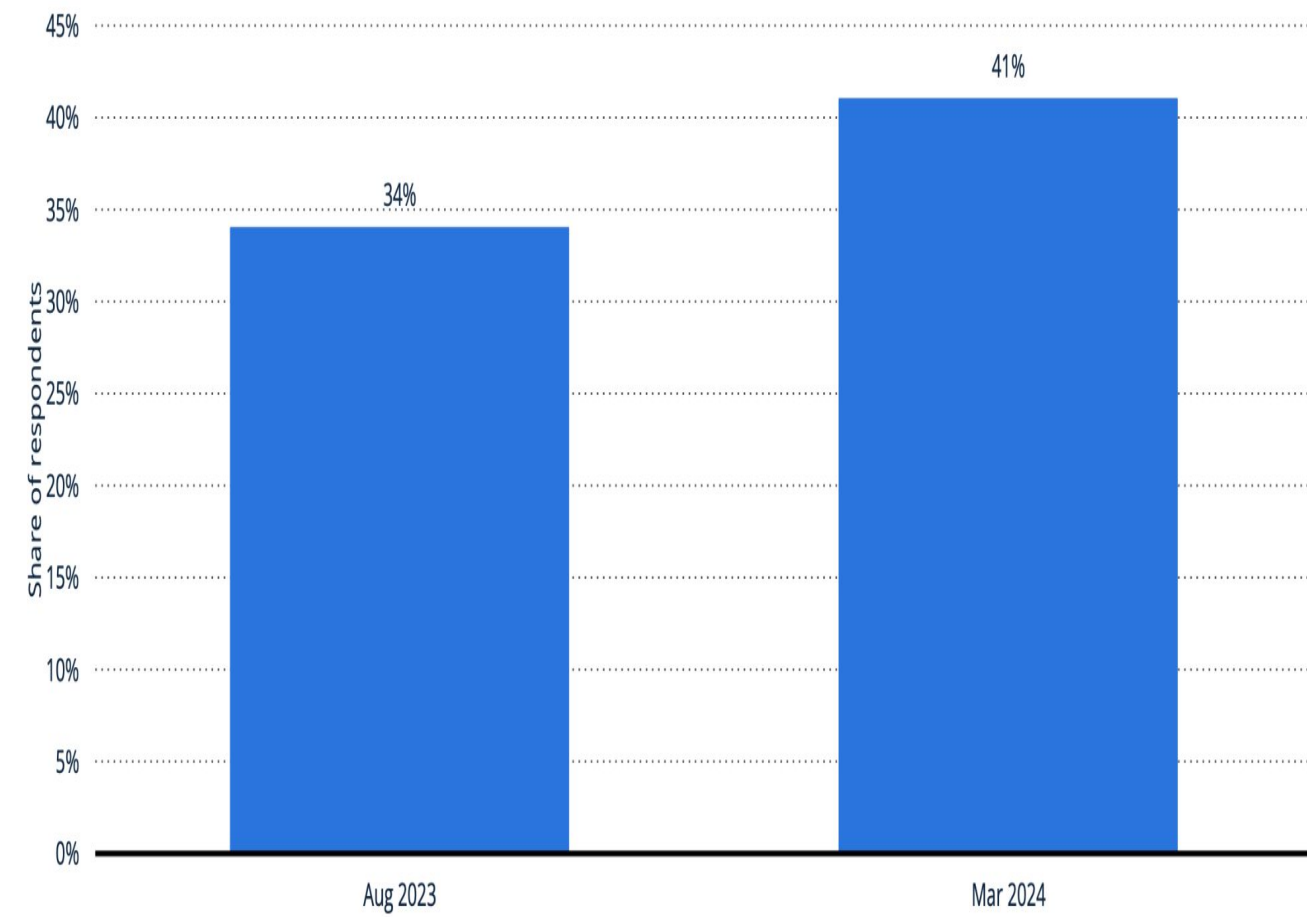
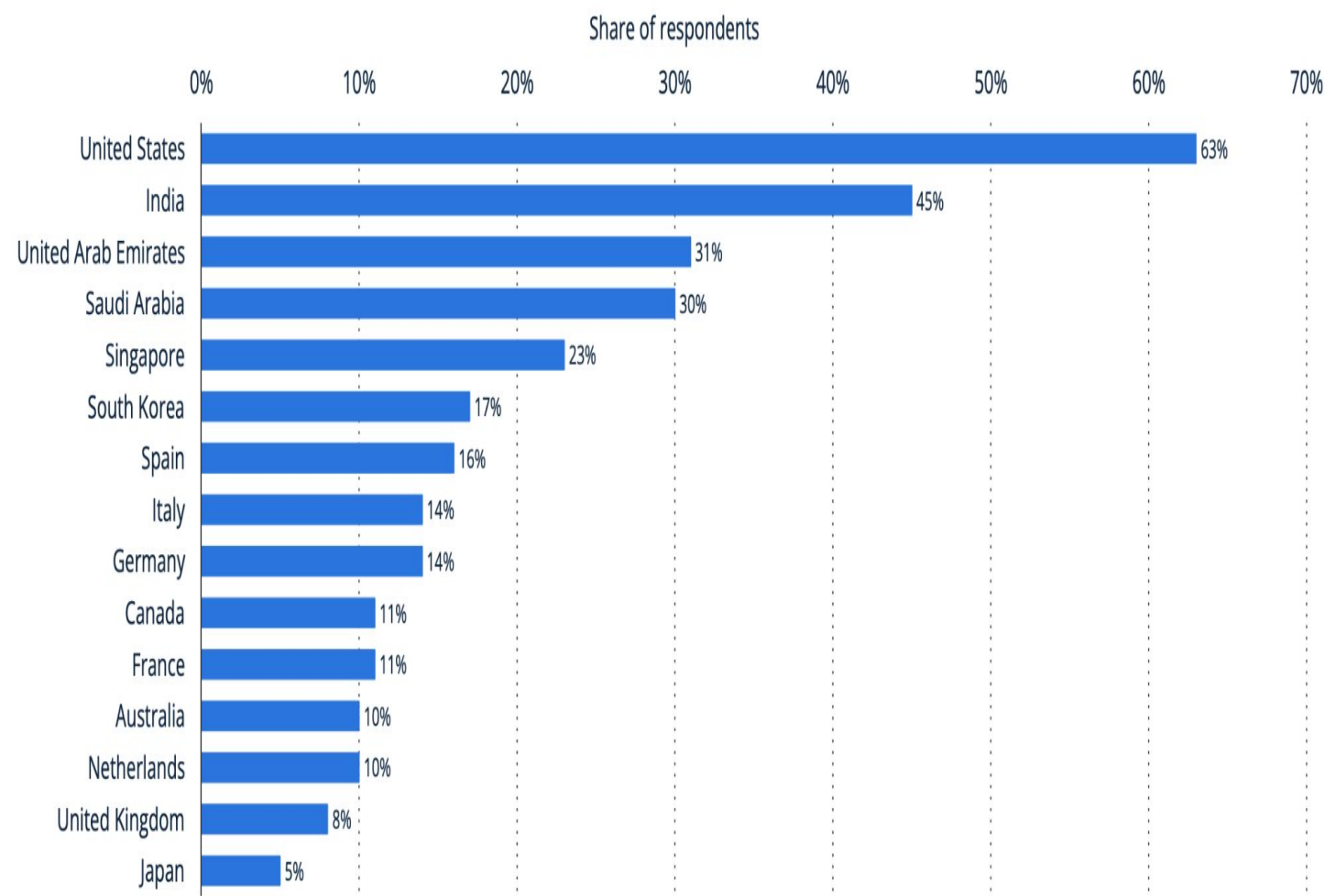
Thomas Cook can capitalize on the U.S. market's booming domestic and international tourism by leveraging AI-driven solutions. This approach positions the company to meet growing demand while staying competitive in a tech-driven, high-potential market.

Growth in Utilization of AI in various countries compared to USA



Share of travelers who research travel with an AI chatbot worldwide as of October 2023

Share of travelers who recently used GenAI for travel inspiration and/or itinerary planning in the United States as of Aug 2023 and March 2024

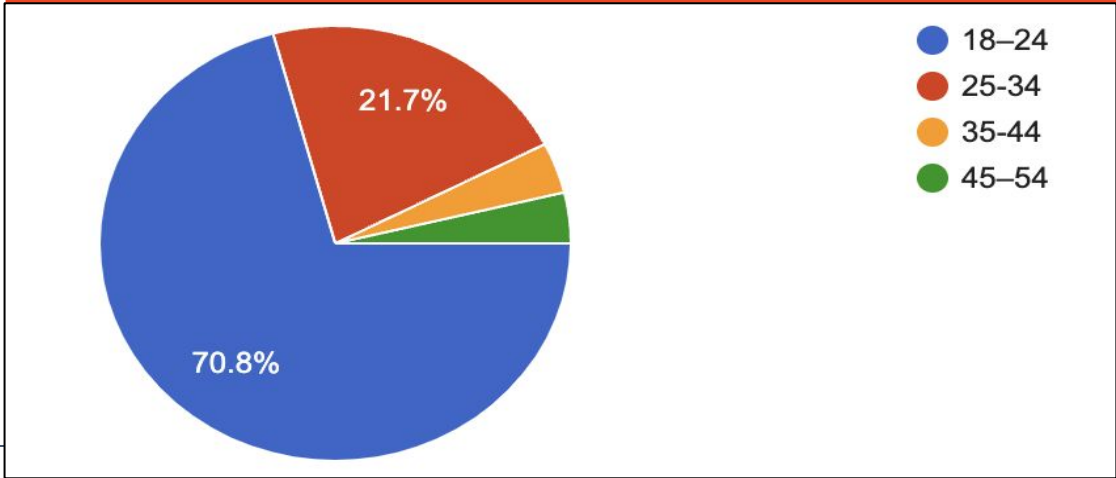


The increasing use of AI in travel planning suggests a rising consumer movement toward digital tools. Thomas Cook should implement such technologies to attract the major chunk of customer base.

Trend Analysis among 106 Travellers

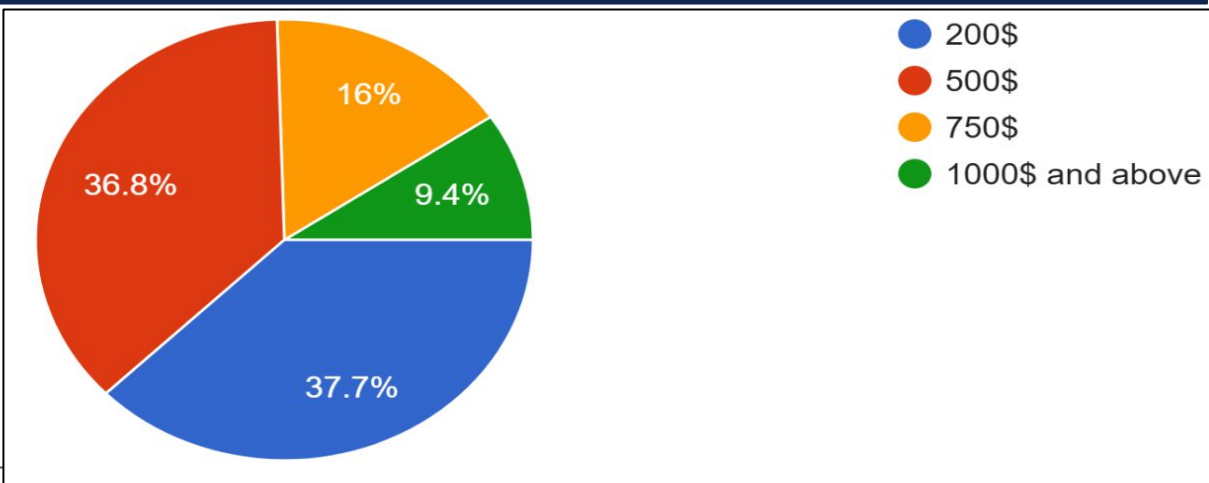


Age Distribution of Travelers



The majority (70.8%) of travelers are aged 18-24, indicating a **predominantly younger audience**.

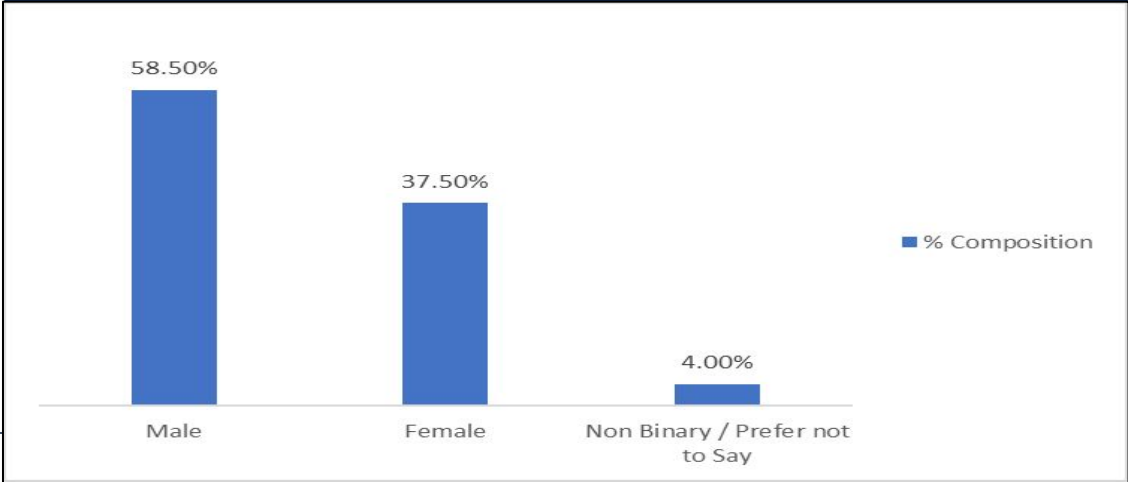
Travel Budget Preferences Per Capita



Most travelers (36.8%) prefer a **budget range of below \$500**

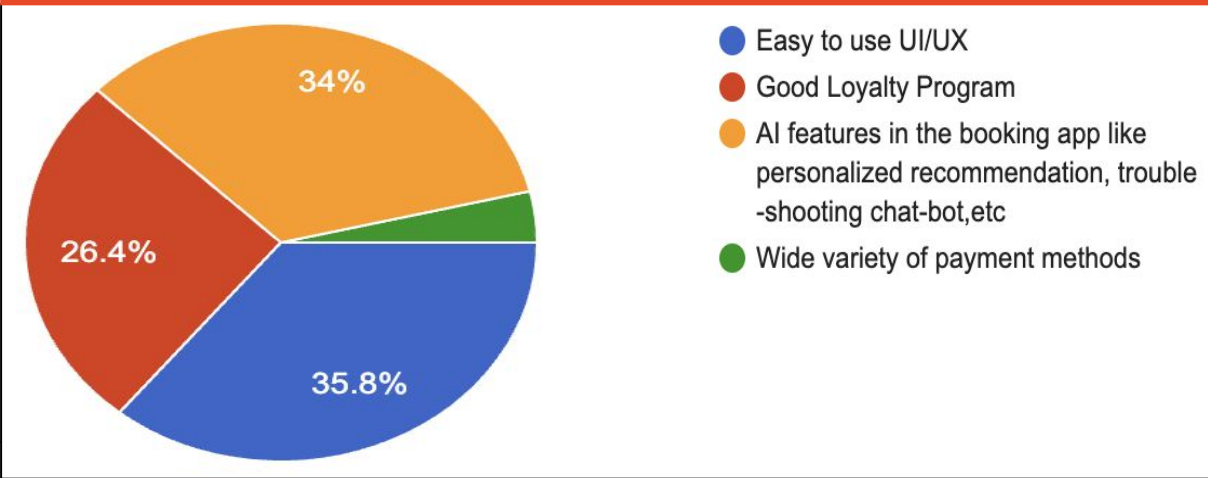


Gender Representation



The **majority of travelers are male (58.5%)**

Factors Influencing Travel Platform Choice



Easy-to-use UI/UX (35.8%) and **AI Features (34%)** are the most significant factors influencing travelers' choice of platforms.

Trend Analysis among 106 Travellers

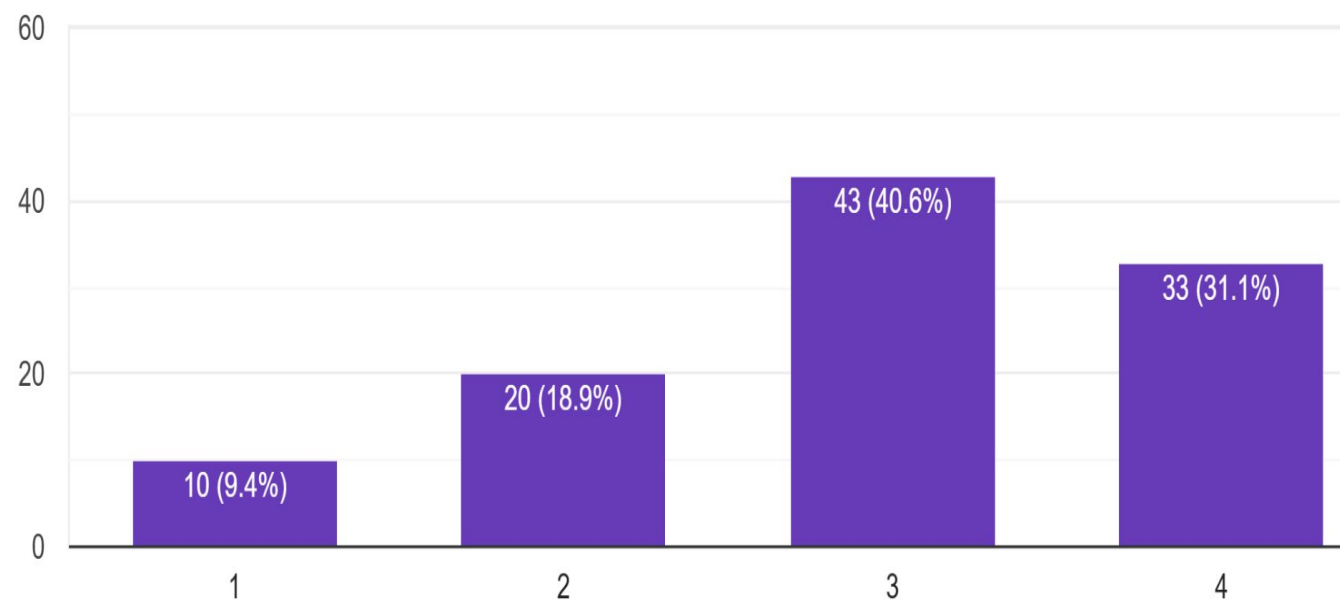


Interest in AI-Powered Itinerary Assistance

Over 70% of respondents showed strong interest in AI-powered travel itinerary services.

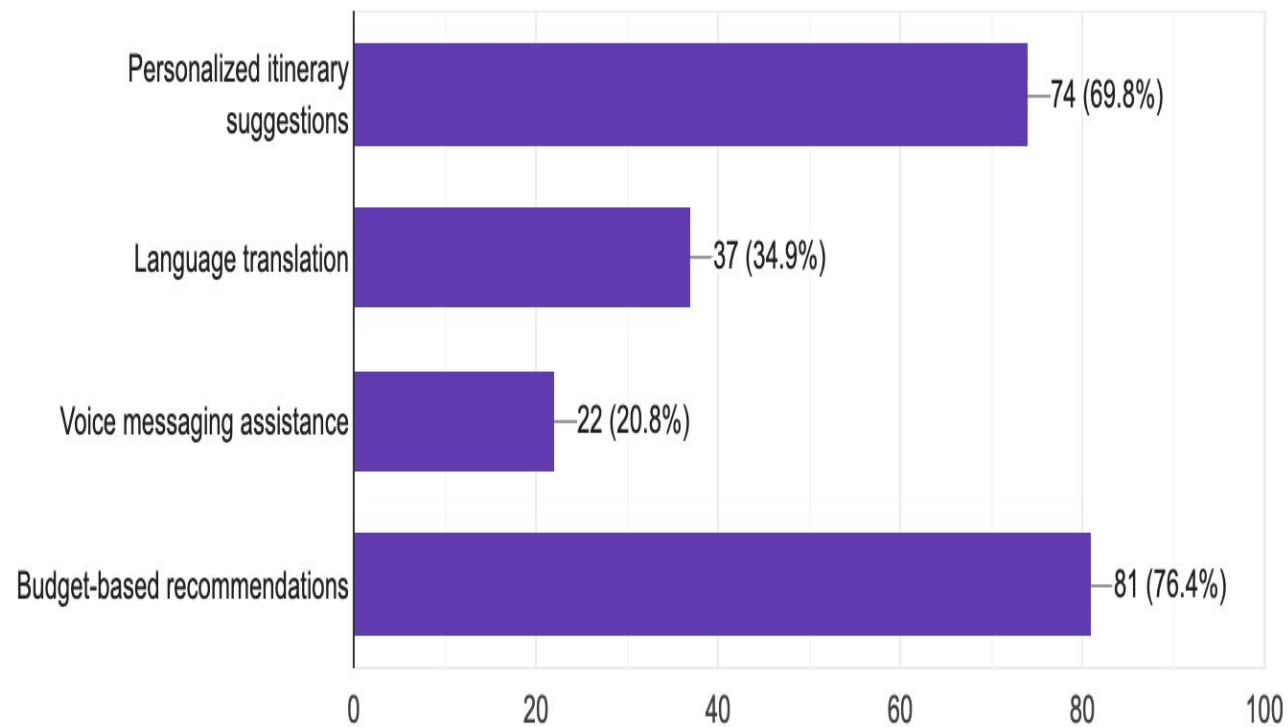
On a scale of 1-4, how likely are you to use a travel service with an AI-powered itinerary assistant?

106 responses



Travel Preferences

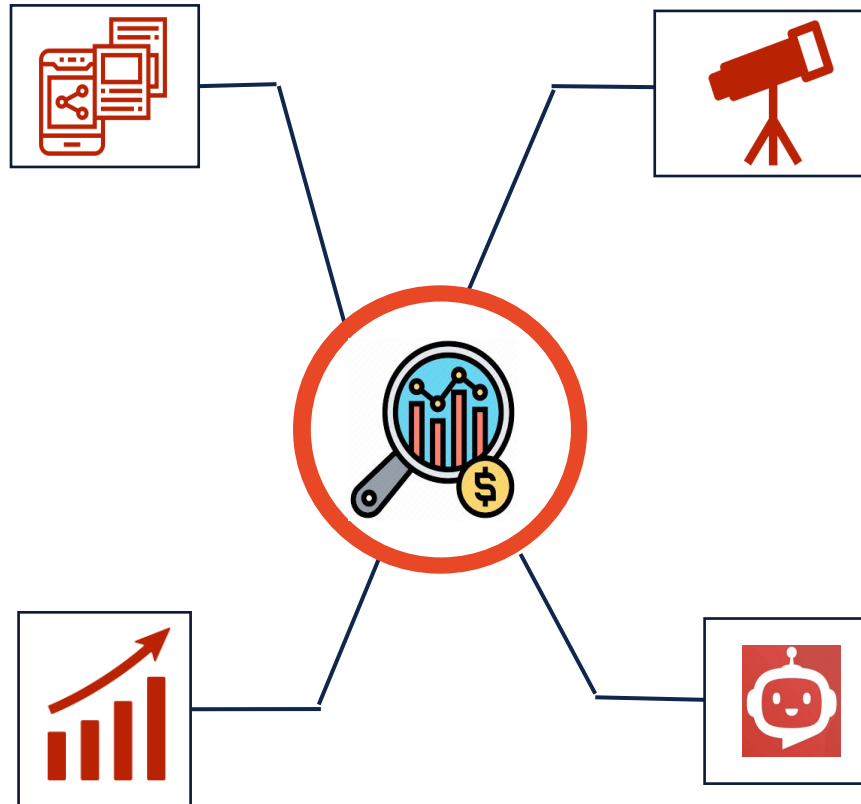
Budget based recommendation system was the top choice of the demographic, Personalized itinerary suggestions being a close second.



With strong demand for AI among travellers, Thomas Cook can acquire a prominent market share if we address the above needs of travellers

Market Research and Analysis

22% of travelers globally now rely on **AI technologies** for trip planning and research.



The **US's tourist market** are forecasted to grow by **20.5% y-o-y** in **2024**, reaching **80.1 million**, surpassing pre-pandemic levels.

Generative AI has the potential to add **\$170–\$300 billion** annually to the travel and tourism sector, equating to **1.2–2%** of the industry's annual revenue.

63% of U.S. travelers use **AI chatbots** on mobile devices for trip planning

Thomas Cook can redefine the U.S. travel market by harnessing AI-driven technologies and tapping into the growing demand for innovative personalized travel experiences

What are Competitors doing ?



1

Expedia



- **Romie** an AI-powered travel assistant using large language models.
- Assist with personalized trip planning, itinerary suggestions and real time travel support.
- Romie is in **beta testing** currently.

Challenges:

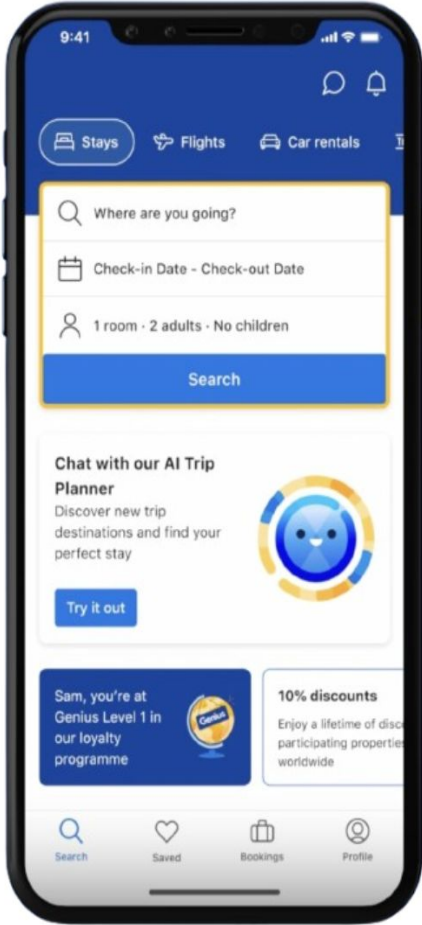
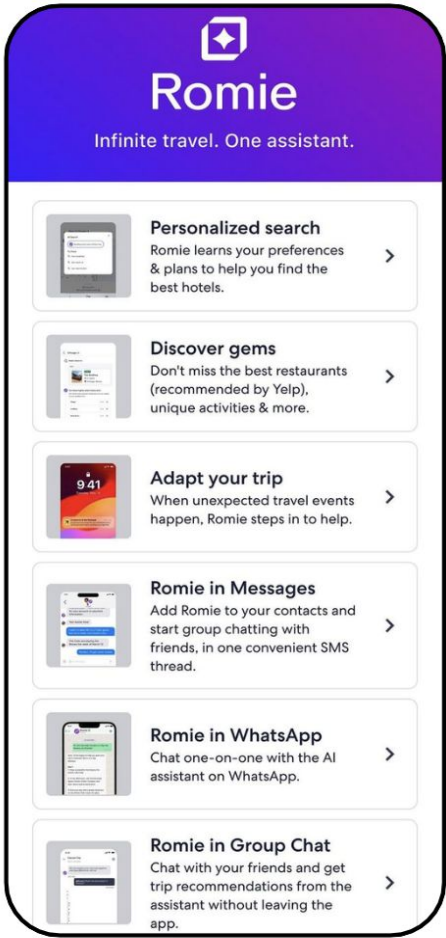
- Early-stage AI tool with limited impact on conversions.
- User interaction with AI is still evolving; learning what resonates with customers.

2

Booking.com



- Booking.com's **Connected Trip Generative AI** uses generative AI for dynamic, real-time recommendations based on location and preferences.
- Unified Travel Platform where users manage entire trips, from booking to activities.



To rival Expedia and Booking.com, Thomas Cook must prioritize significant investments in AI-driven chatbots and personalized, budget-friendly travel recommendations

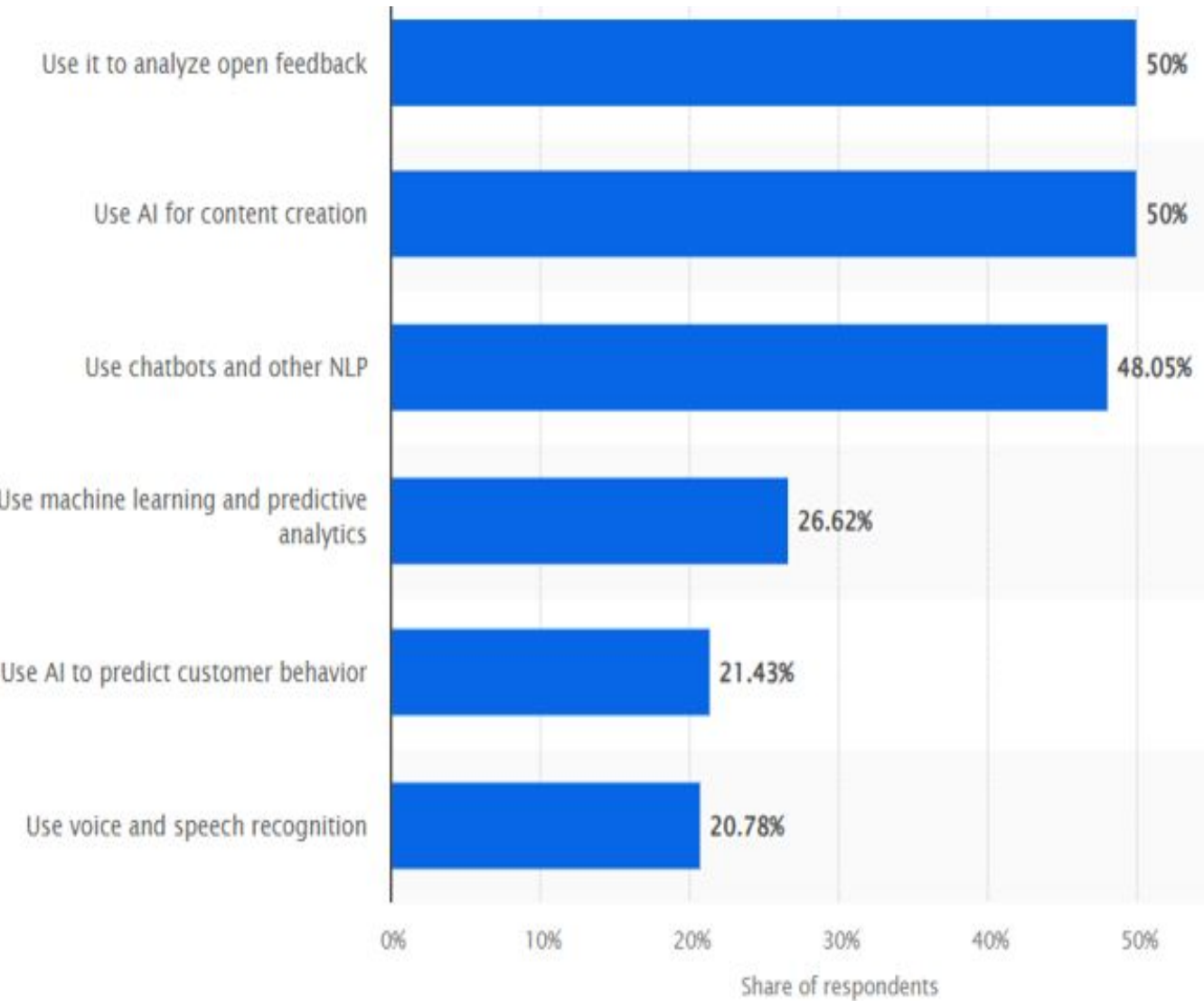


Strategies to integrate Artificial Intelligence

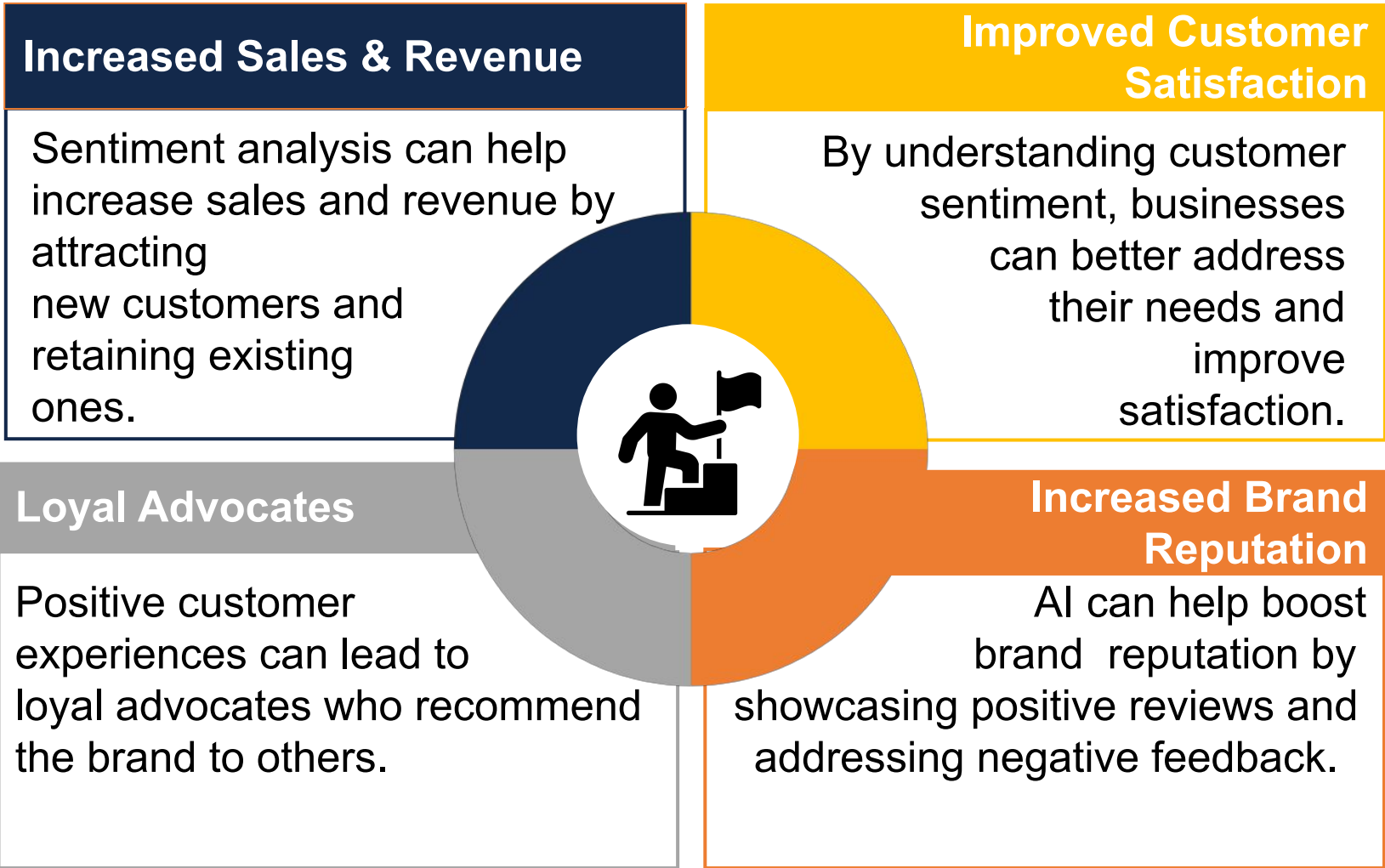
Customer Sentiment Analysis



Survey of Worldwide 2024: 10,000
Decision Makers



Key Components driving Success



By leveraging sentiment analysis, Thomas Cook can not only improve its operations and marketing but also strengthen its reputation and customer loyalty in a highly competitive industry.

Transforming Travel with Personalized Itineraries



Customers preferences

- **Personalized itinerary suggestions** are a top feature for younger users in terms of time savings and convenience.
- **Millennials** who have been motivated by social media are looking for tailored, dynamic travel itineraries that are relevant to their interests.
- **Voice assist and language translation** are valued by older populations as complementary capabilities to a single platform.



Personalized Itinerary

- **Integrated Options:** One-stop programs adapted to users' needs, eliminating the need for multiple bookings.
- **Real-Time Updates:** Flexible itineraries that react quickly to disruptions or changes.
- **Customizability:** Travel plans are easy to alter and include recommendations for add-ons such as tours and dining experiences.

Building Loyalty: One Itinerary at a Time, This technology not only simplifies the booking process, but it also increases trust, loyalty, and happiness, establishing Thomas Cook as the go-to platform for seamless travel.

Elevating Travel Experiences with General AI Enhancements



Customer Expectation	- Travelers value real-time assistance in answering questions, adjusting , itineraries, and providing personalized support. - Millennials favor conversational AI technologies for rapid solutions, but older populations value user-friendly interfaces with multilingual assistance.
Solution	Improve Customer Support: Provide 24-7 conversational support with trip planning, FAQs, and real-time problem solving. Multilingual Accessibility: Allow for smooth communication in multiple languages to reach a worldwide audience. Interactive Itinerary Adjustments: Allow customers to adjust itineraries via speech cues, increasing flexibility and ease.

Thomas Cook's integration of human-like conversational AI provides better support, bespoke travel solutions, and a seamless planning experience—all on one platform.

Predictive Analytics for Demand Forecasting



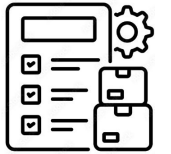
Functions

Predicts when and where customer demand will peak or dip, enabling better decision-making for inventory, staffing, and marketing efforts.

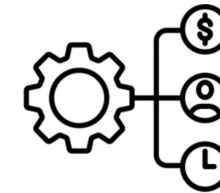
Technology

Uses **Time Series Analysis** to analyze and forecast data points collected over time. In demand forecasting, it identifies patterns such as seasonality, trends, and anomalies, helping predict future demand for services like flights, hotels, or travel packages.

Accurate Inventory Management



Optimized Resource Allocation



Enhanced Customer Experience



Cost Efficiency



Key Benefits

Predictive demand forecasting empowers Thomas Cook to maximize resource utilization, enhance promotional planning, and mitigate booking risks for seamless customer experiences.

Dynamic Pricing and Optimization



Functions

Determines optimal prices that balance profitability and competitiveness, adapting instantly to market changes such as peak travel seasons or off-peak discounts.

Technology

Leverages **Reinforcement Learning** to dynamically adjust prices based on factors like demand, competitor pricing, and customer behavior in real-time, enabling firm to stay agile in a competitive market and enhance profitability through data-driven decision-making.

Maximized Revenue



Enhanced Customer Experience



Market Competitiveness



Increased Booking Engagement



Key Benefits

With this Thomas Cook can maximize revenue during high-demand periods and fill gaps during low-demand times, while offering competitive prices to customers.

Predictive Emergency and Health Alerts



	Personalized Health Insights	Emergency Alerts and Assistance	Seamless Integration in the App
Functions	Tracks health metrics, provides tips, sends alerts for risks, and connects to telemedicine.	Sends location-based safety alerts, suggests alternate plans, and enables one-tap SOS.	Offers customized alerts, offline access, and automated support for a worry-free experience.
Technology	Wearable integration (Apple HealthKit, Google Fit), AI for analysis (TensorFlow), telemedicine APIs.	GPS tracking (Google Maps), risk analysis (IBM Weather API), emergency services APIs.	Machine learning for preferences, push notifications (Firebase), offline data caching
Benefits	Keeps travelers healthy, reduces stress, and boosts trust in the brand.	Enhances safety, minimizes disruptions, and reinforces Thomas Cook's reliability.	Personalizes the experience, works in low-connectivity areas, and builds loyalty.

Incorporating predictive health and emergency alerts will transform Thomas Cook into a reliable partner, prioritizing customer well-being while fostering trust and satisfaction.



Legal Barriers to Entry in the United States Market

Legal Barriers to Entry: Licensing, Registration, and Financial Security



Legal Requirements

Thomas Cook will need to register with the correct local and state authorities and obtain the proper licenses for selling travel to a location or selling travel in a location. Some states have different legal requirements than others, especially California, Florida, Hawaii, Iowa, and Washington. States may require business insurance as well.



International Air Transport Association (IATA)

IATA Travel Agent accreditation will simplify any business with airlines and other industry partners. This includes airline bookings, access to facilities, payment flexibility, and credibility. Being accredited may bring more business and trust to Thomas Cook.



Financial Security

Thomas Cook may need to prove financial security, depending on location. To do so, Thomas Cook will need to purchase liability insurance or a bond to ensure coverage of certain customer situations. Some states will accept a letter of credit, certificate of deposit, or trust account as proof of financial security.

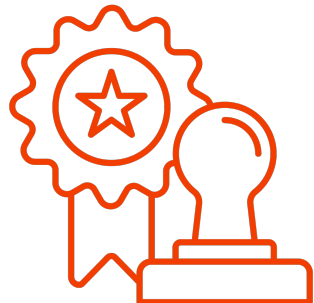
Thomas Cook will need to follow state regulations for registration and licensing in order to legally sell travel in a location or to a location, and for proving financial security. IATA accreditation will add credibility and ease to Thomas Cook operations.

Legal Barriers to Entry: Taxes and Intellectual Property



Tax Regulations

The U.S. Supreme Court decision in *South Dakota v. Wayfair* allows states to collect taxes from sellers that are not physically located in the state, which includes online sales. Therefore, Thomas Cook will have to collect sales tax and pay taxes in each state where an online sale occurs.



Trademarks

Thomas Cook will need to register any trademarks with the U.S. Patent and Trademark Office (USPTO). A trademark is a word, phrase, symbol, design, or combination that identifies your goods or services. Thomas Cook must submit an application and get it approved before the trademark is registered. Afterwards, the trademark must be maintained in order to keep it registered.

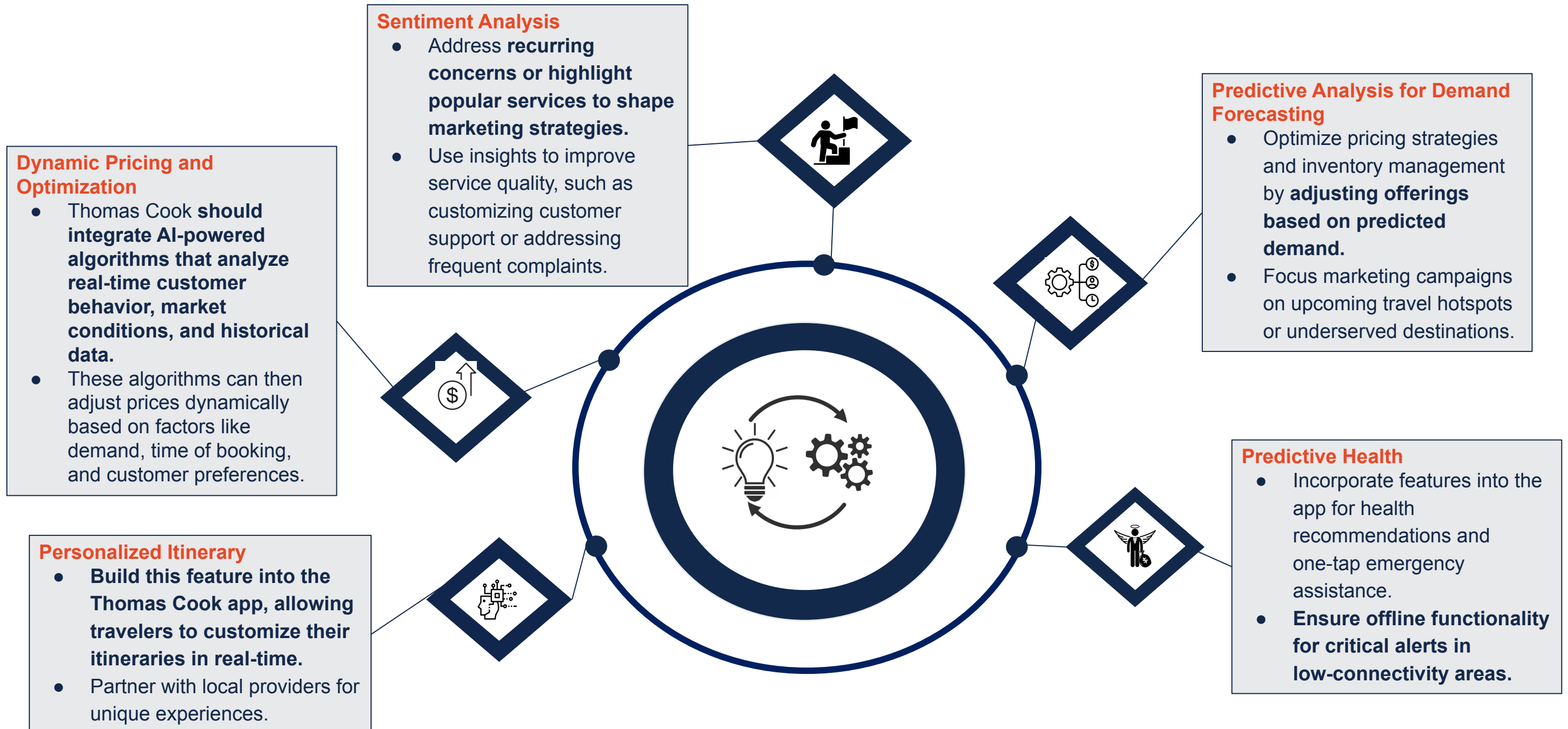


Registration Benefits

Registering a trademark provides legal protections for branding, protects against counterfeiting and fraud, and allows control of how the trademark is used in the industry.

Thomas Cook will have to collect and pay sales tax in every state that a sale occurs. Thomas Cook will also need to register any trademarks with the USPTO to ensure protections for branding and to retain control over how the branding is used in the travel industry.

How to proceed with the implementation?



Q&A



Why Thomas Cook and not Expedia ?



1

Expedia

- Expedia is a strong competitor with advanced AI infrastructure in testing; nonetheless, the opportunity for significant innovation is restricted. Expedia already operates at a high level of optimization.
- The level of collaboration and influence is not as feasible with a mature, market-leading organization like Expedia.

Challenges:

Making Thomas Cook enter the US Market and collecting the data from US travel industry to compare it to that of Thomas Cook's data was a bit challenging. We studied about government barriers and regulations to tackle this issue.

2

Thomas Cook

- Thomas Cook provides a more dynamic challenge and more opportunities to leverage our skills in consulting, data analytics, and AI-driven solutions to achieve demonstrable results.
- Thomas Cook has allowed us to play an important part in its efforts to modernize its operations and reestablish itself as a premier travel brand.



Thomas Cook is a client where our efforts can have a genuine impact, both in terms of increasing their market position and driving substantial innovation in the travel industry.



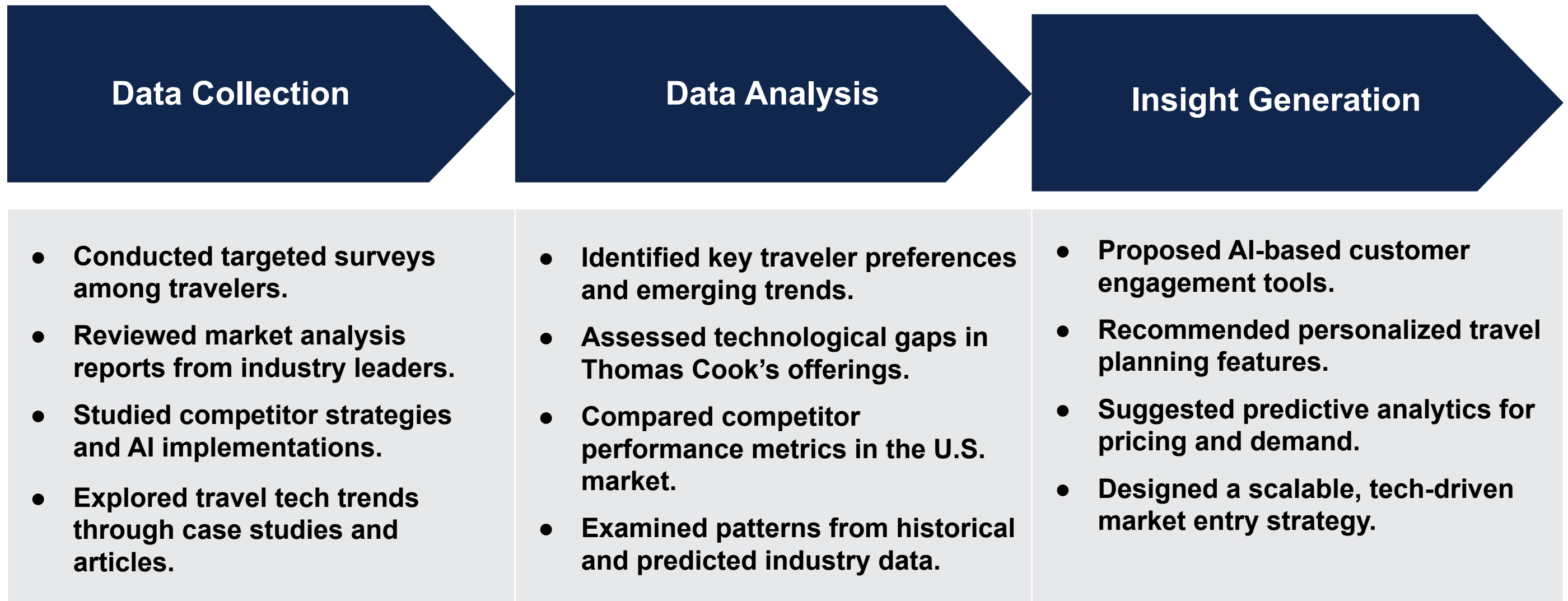


Appendix

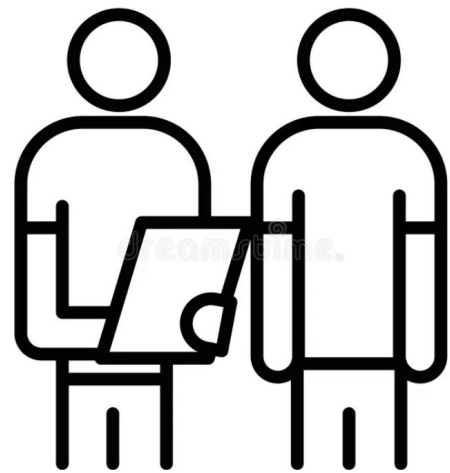


How We Conducted Research: Inductive Approach

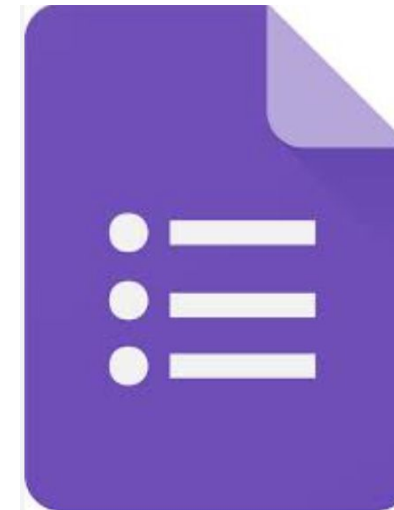
It is a method where we started with specific observations and data, analyzed patterns within them, and then developed theories or hypotheses based on those patterns and results of the survey.



Primary Research Survey



Our team conducted *a survey with 106 participants on Oct 27th, 2024*, exploring topics such as age, gender, travel preferences, budget considerations, and the role of Artificial Intelligence in the travel industry.



The survey, created using **Google Forms**, primarily targeted individuals within the *median age range of 18–24*.

Why did we choose the 18-24 age group?



Early Adopters of Technology

They are **more open to using AI-powered tools** and digital services for travel planning.

Emerging Independent Travelers:

Many in this age group are **experiencing independent travel for the first time**, making them a key audience for understanding preferences and trends.



Budget-Conscious and Experience-Driven:

They **prioritize value for money** and unique experiences, making them a prime audience for innovative travel solutions.

Potential Long-Term Customers

Insights into their behavior can help shape products and services that foster loyalty as they **continue to travel in the future**.



Key Questions from our Survey



What is the single most important thing you consider before using a platform for planning your trip? *

- ☐ Easy to use UI/UX
- ☐ Good Loyalty Program
- ☐ AI features in the booking app like personalized recommendation, trouble -shooting chat-bot,etc
- ☐ Wide variety of payment methods

Likelihood to Use AI-Powered Itinerary Services:

Determines the potential adoption of AI technology in travel planning, helping gauge its market appeal.

What is your typical budget range for travel? *

- ☐ 200\$
- ☐ 500\$
- ☐ 750\$
- ☐ 1000\$ and above

Travel Budget Insights:

Helps identify the typical spending range of travelers, enabling targeted service offerings and pricing strategies.

Key Questions from our Survey

On a scale of 1-4, how likely are you to use a travel service with an AI-powered itinerary assistant?

1

2

3

4

☐
☐
☐
☐

Relevance of AI-Powered Itinerary Assistants:

Understanding user willingness to adopt AI-powered services helps gauge the potential market demand and refine service offerings.

Which of these AI features would you find most helpful? (Select up to 2)

- ☐ Personalized itinerary suggestions
- ☐ Language translation
- ☐ Voice messaging assistance
- ☐ Budget-based recommendations

Helpful AI Features:

Identifying which AI features resonate most with users allows for targeted development of tools that enhance travel experiences effectively.

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Thank You !

